

20.25

Second Mid Sem Exam- BIO202-Marks-25-Duration 1hr

-Name: SIDDHANT PANDA

-Roll No: MS24129

Questions 1-30 carry 0.5 marks and 0.25 negative marking for wrong answers

12.5 - 1.25 =
11.25

1. Which cellular organelle has its own DNA and ribosomes, indicating an endosymbiotic origin?

- A. Golgi apparatus
- ☒ B. Mitochondrion
- C. Peroxisome
- D. Lysosome

2. In plant cells, which organelle is analogous in function to the lysosome in animal cells?

- A. Central vacuole
- ☒ B. Peroxisome
- C. Plastid
- D. Ribosome

3. Which of the following statements about G proteins is *true*?

- A. G proteins are permanently bound to GTP.
- B. The active form of a G protein has GDP bound to it.
- C. The α subunit is inactive while bound to GTP.
- ☒ D. GTP hydrolysis inactivates the G protein, returning it to the receptor.

4. If a cell has 20 chromatids, how many centromeres are present?

- A. 30
- B. 20
- ☒ C. 10
- D. 40

5. The peroxisome differs from the lysosome mainly because it:

- ☒ A. Breaks down fatty acids via β -oxidation
- B. Participates in protein synthesis
- C. Contains DNA
- D. Is found only in prokaryotes

6. The main site of glycosylation (addition of carbohydrate chains) to proteins is:

- ☒ A. Rough endoplasmic reticulum
- B. Ribosome
- C. Smooth endoplasmic reticulum
- D. Golgi apparatus

7. In animal cells, at which mitotic phase do centrioles begin to move apart?

- A. Telophase
- B. Anaphase
- ☒ C. Prophase
- D. Prometaphase

8. Which of the following best describes the evolutionary relationship between binary fission and mitosis?

- A. Both evolved independently from viral replication
- B. Binary fission evolved from eukaryotic mitosis
- C. They are unrelated processes
- ☒ D. Mitosis evolved from binary fission in ancestral prokaryotes

9. Which of the following statements about the endomembrane system is FALSE?

- A. The nuclear envelope is considered part of the endomembrane system.
- B. The Golgi apparatus and ER membranes are in continuity via vesicle trafficking.
- ☒ C. Mitochondria are part of the endomembrane system because they have inner and outer.
- D. Lysosomes originate from Golgi-derived vesicles that contain hydrolytic enzymes.

10. What is the role of non-kinetochore (i.e. interpolar) microtubules during mitosis?

- A. They shorten to move chromosomes
- B. They attach to centromeres
- ☒ C. They maintain appropriate overlap of microtubules in the cell's center
- D. They disassemble completely during telophase

11. A protein is synthesized on the rough ER, folded, and glycosylated. It is then transported in a vesicle to the Golgi, modified further, and finally secreted. Which *sequence* of compartments best describes this pathway?

A. Rough ER → Lysosome → Golgi → Plasma membrane

☒ B. Rough ER → Golgi → Transport vesicle → Plasma membrane

C. Golgi → Rough ER → Secretory vesicle → Plasma membrane

D. Rough ER → Transport vesicle → Endosome → Plasma membrane

12. The term "generation time" in bacterial growth refers to:

A. Time required for DNA replication

☒ B. Time between successive cell divisions

C. Time needed for colony formation

D. Time for mutation and adaptation

13. In signal transduction, what is the immediate effect of a signaling molecule binding to a receptor protein?

A. Activation of gene transcription

☒ B. A conformational change in the receptor

C. Production of second messengers

D. Transport of the receptor to the nucleus

14. Suppose a cell is actively secreting digestive enzymes. Which organelle(s) would you expect to be particularly well developed or abundant in that cell?

A. Smooth ER and peroxisomes ✗

B. Lysosomes and mitochondria

☒ C. Rough ER and Golgi apparatus

D. Chloroplasts and vacuoles ✗

15. What causes the decrease in cyclin concentration at the end of M phase (mitosis)?

A. Increase in cyclin production

☒ B. Its destruction triggered by its complex with Cdk

C. Changing ratio of cytoplasm to genome

D. Its binding to chromatin

16. A sub-lethal mutation disrupts the nuclear lamina (a network of protein filaments on the inner side of the nuclear envelope). What cellular outcome is most likely?

☒ A. Inability to transport RNA out of the nucleus

B. Loss of the nucleus entirely

C. Altered shape of the nucleus

D. Immediate cell death

17. Which of the following correctly matches the messenger type with its typical distance of action?

☒ A. Autocrine — acts on the same cell that secretes it

B. Paracrine — long distance through bloodstream

C. Endocrine — short distance, local signaling

D. Synaptic — acts on cells in distant tissues

18. Which statement about the smooth ER is TRUE?

A. It has ribosomes attached.

B. It is continuous with the inner mitochondrial membrane.

☒ C. It glycosylates proteins destined for secretion.

D. It stores calcium ions and is involved in detoxification.

19. G protein-coupled receptors (GPCRs) are characterized by:

☒ A. Seven transmembrane α -helices and activation of GTP-binding proteins

B. Intracellular kinase activity

C. Single transmembrane α -helix structure

D. Direct binding to DNA to alter transcription

20. The structure that forms during bacterial cell division to separate daughter cells is called:

A. Septum

☒ B. Cleavage furrow

C. Cell plate

D. Fission bridge

21. In cell signaling, the term "ligand" refers to:

- A. A small molecule that inhibits transcription
- B. A transcription factor in the nucleus
- C. A protein that phosphorylates another protein
- ☒ D. A molecule that specifically binds to a receptor protein

22. During binary fission, the origin of replication:

- A. Remains fixed at the cell's midpoint
- B. Fuses with the nucleoid of the daughter cell
- C. Is degraded before cytokinesis
- ☒ D. Is attached to the plasma membrane and moves as the DNA replicates

23. Which of the following signaling types involves direct contact between cells?

- ☒ A. Paracrine signaling
- ☒ B. Juxtacrine signaling
- C. Synaptic signaling
- D. Endocrine signaling

24. Which statement about signal amplification is true?

- A. It requires multiple different ligands
- B. It always decreases energy use
- C. Amplification occurs only in nuclear signaling
- ☒ D. A single activated kinase can phosphorylate many targets

25. Which checkpoint ensures that all chromosomes are attached to the spindle before separation?

- A. G₁ checkpoint
- B. G₂ checkpoint
- ☒ C. M checkpoint (spindle checkpoint)
- D. Cytokinesis checkpoint

26. Why is the surface area-to-volume ratio (SA:V) an important limiting factor for cell size, especially in eukaryotic cells?

- A. Because diffusion becomes too fast in large cells.
- ☒ B. Because as a cell's volume increases, the relative surface area decreases, limiting exchange of materials.
- C. Because large cells can't maintain a cytoskeleton.
- D. Because DNA replication is hindered in larger cells.

27. Binary fission in bacteria differs from mitosis in that:

- A. It involves the breakdown of the nuclear envelope
- ☒ B. It does not involve spindle formation or mitotic phases
- C. It produces genetically diverse daughter cells
- D. It requires microtubules to separate chromosomes

28. Which of the following enzymes removes phosphate groups from proteins, thus inactivating them?

- ☒ A. Phosphatase
- B. Adenyl cyclase
- C. Protein kinase
- D. GTPase

29. Which stage of signaling involves amplification of the original extracellular signal?

- A. Reception
- ☒ B. Transduction
- C. Response
- D. Termination

30. The mitotic spindle is primarily made of:

- A. Intermediate filaments
- B. Actin filaments
- ☒ C. Microtubules composed of tubulin
- D. Microfibrils of cellulose

State True or False. Each question carries 0.5 marks. No negative marking

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1	Signal transduction pathways are isolated and do not interact with one another.	True	False
2	Ribosomes can be found both freely in the cytoplasm and attached to the rough endoplasmic reticulum.	True	False
3	A mutation that prevents GTP hydrolysis in a G protein would cause continuous signal activation.	True	False
4	Binary fission occurs in prokaryotic cells and involves spindle formation.	True	False
5	During metaphase, chromosomes line up at the equator of the cell known as the metaphase plate.	True	False
6	Signal transduction cascades often include both protein kinases and phosphatases to allow precise control.	True	False
7	The nuclear envelope completely disappears during metaphase.	True	False
8	The Golgi apparatus modifies and sorts proteins before they are sent to their final destinations.	True	False
9	In cell signaling, the ligand always enters the target cell to trigger a response.	True	False
10	Receptor tyrosine kinases (RTKs) function as dimers and can activate multiple intracellular pathways simultaneously.	True	False
11	Chloroplasts are found in all eukaryotic cells.	True	False
12	Signal termination is as essential as activation in maintaining cellular homeostasis.	True	False
13	The rough endoplasmic reticulum (RER) is the site of lipid synthesis.	True	False
14	The cytoskeleton is made only of microtubules.	True	False
15	The mitotic spindle is composed of microtubules that originate from centrosomes in both plant and animal cells.	True	False
16	Protein kinases transfer phosphate groups from ATP to target proteins.	True	False
17	In eukaryotes, spindle fibers attach to chromosomes at the kinetochore.	True	False
18	The Golgi apparatus receives proteins from the ER, modifies them, and ships them in vesicles.	True	False
19	Cytokinesis is part of mitosis.	True	False
20	G protein-coupled receptors (GPCRs) activate intracellular G proteins by exchanging GDP for GTP.	True	False